

**Report
Of
Summer Internship Project at Busung Tape & Film India
Private Limited**

Submitted

To

**School of Computer Science and Engineering
IILM University, Greater Noida, U.P.**

**In partial fulfilment of the requirement of degree of
B.Tech CSE – AI & ML**



By

Name of Student: ANURAJ KUMAR

Roll Number: 25SCS1003001412

Batch: 2CSE7

Company: Busung Tape & Film India Private Limited

Internship Duration: 20 June 2026 – 10 July 2026

CANDIDATE'S DECLARATION

I, Anuraj Kumar, student of B.Tech (Computer Science and Engineering – Artificial Intelligence and Machine Learning), Roll No. 25SCS1003001412, Batch 2CSE7, School of Computer Science and Engineering, IILM University, Greater Noida, hereby declare that the Internship Report entitled “Internship at Busung Tape & Film India Private Limited” submitted in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology is a bona-fide record of the work carried out by me during the period from 20 June 2026 to 10 July 2026.

I further declare that this report has not been submitted to any other University or Institution for the award of any degree, diploma, certificate or other similar title. All information presented herein is true to the best of my knowledge and is based on actual work performed during the internship under the guidance of the training supervisors at Busung Tape & Film India Private Limited.

I understand that any false statement or misrepresentation of facts in this report may lead to appropriate disciplinary action by the University.

Date: _____

Place: Greater Noida

Name of Student: Anuraj Kumar

Roll Number: 25SCS1003001412

Batch: 2CSE7

Programme: B.Tech (CSE – AI & ML)

Signature of Student: _____

BS
SLITTING
LAMINATING
COATING

Busung Tape & Film India Private Limited

CIN: (U74999UP2019FTC115758)

F-5, SITE 5, UPSIDA, KASNA, SURAJPUR, GREATER NOIDA, UTTAR PRADESH, INDIA-201306

EMAIL: busung.india@gmail.com TEL: 0120-2341192

INTERNSHIP COMPLETION CERTIFICATE

Certificate No.: BTC/INT/2026/ 02

Date: 10 July 2026

TO WHOMSOEVER IT MAY CONCERN

This is to certify that Mr. Anuraj Kumar
Roll No. 25SCS1003001412, a B.Tech (CSE – AI & ML) student of
IILM University, Greater Noida, has successfully completed his Paid Internship
at **BUSUNG TAPE & FILM INDIA PRIVATE LIMITED** from
20 June 2026 to 10 July 2026.

During the internship, he worked in the following areas:

- Sales & Marketing
- Data Analysis & Reporting
- Microsoft Excel Operations
- Business Data Management

Throughout the internship, Mr. Anuraj Kumar
demonstrated dedication, sincerity, willingness to learn, and professionalism.
His performance was found to be satisfactory, and we appreciate his
contribution to the organization.

We wish him every success in his future academic and professional endeavors.

For

BUSUNG TAPE & FILM INDIA PRIVATE LIMITED

Anuraj Kumar
BUSUNG TAPE & FILM INDIA PVT. LTD.

Authorized Signatory

(HR/Director) Authorised Signatory



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ACKNOWLEDGMENT

I take this opportunity to express my deep sense of gratitude to Busung Tape & Film India Private Limited for providing me with a valuable opportunity to undergo a paid internship in their esteemed organization. The practical exposure and real-world problem-solving experience gained during these three weeks have significantly enriched my academic learning.

I am especially thankful to Mr. Avinash Kumar, Director of the company (DIN: 08488139), for his guidance, encouragement and for creating an environment in which interns are allowed to observe, analyse and contribute to actual production and business processes. His support made it possible for me to work on a meaningful process-improvement activity related to material wastage reduction.

I extend my sincere thanks to the entire team of Sales & Marketing, Production and Administrative staff who patiently explained the company processes, shared relevant data and allowed me to participate in day-to-day operations. Their cooperation was instrumental in the successful completion of the internship.

I am grateful to the faculty members of the School of Computer Science and Engineering, IILM University, Greater Noida, for their continuous academic guidance and for designing a curriculum that prepared me to approach industrial problems with an analytical mindset. Special thanks to the Training and Placement cell for facilitating this internship opportunity.

Finally, I express my heartfelt gratitude to my parents and family members for their unwavering support, motivation and understanding throughout my academic journey and during the internship period.

Any errors or omissions that may remain in this report are solely my responsibility.

Anuraj Kumar

Roll No.: 25SCS1003001412

B.Tech (CSE – AI & ML), Batch 2CSE7

IILM University, Greater Noida

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INTERNSHIP COMPLETION CERTIFICATE (SUMMARY)

(Original certificate issued by the company is to be attached with the hard-bound report)

This is to certify that Mr. Anuraj Kumar, Roll No. 25SCS1003001412, a B.Tech (CSE – AI & ML) student of IILM University, Greater Noida, has successfully completed his Paid Internship at Busung Tape & Film India Private Limited from 20 June 2026 to 10 July 2026.

Certificate No.: BTC/INT/2026/02

Date of Issue: 10 July 2026

Areas of Work during Internship:

- Sales & Marketing
- Data Analysis & Reporting •
- Microsoft Excel Operations •
- Business Data Management

Throughout the internship, Mr. Anuraj Kumar demonstrated dedication, sincerity, willingness to learn and professionalism. His performance was found to be satisfactory, and the company appreciated his contribution to the organization. The company wishes him every success in his future academic and professional endeavors.

Authorized Signatory: Mr. Avinash Kumar (Director / HR), Busung Tape & Film India Private Limited

1. INTRODUCTION

1.1 Background of the Internship

Industrial internships form an essential bridge between theoretical knowledge acquired in the classroom and the practical realities of professional work environments. For a student of Computer Science and Engineering with specialization in Artificial Intelligence and Machine Learning, exposure to real manufacturing and business operations provides invaluable insight into how data, process analysis and systematic thinking can create measurable value for an organization.

I was selected for a paid internship at Busung Tape & Film India Private Limited, a manufacturing company located in Greater Noida, Uttar Pradesh. The internship offer was extended vide letter dated 17 June 2026. The internship period was fixed from 20 June 2026 to 10 July 2026. The position offered was Intern – Sales & Marketing / Data Analysis, with a stipend of 6,000 for the period.

Although the primary designation covered Sales & Marketing, Data Analysis and Microsoft Excel Operations, the company also permitted interns to observe and contribute to production-related improvement activities. This flexibility enabled me to identify a concrete operational problem related to material wastage in the circular die-cutting process and to propose a practical solution that reduced scrap and product cost.

The present report documents the entire internship experience – from the organizational context and assigned responsibilities to the problem analysis, solution implementation, data-related work performed, learning outcomes and recommendations. It is submitted in partial fulfilment of the requirements for the award of the degree of B.Tech (CSE – AI & ML) at IILM University, Greater Noida.

Table 1: Internship Basic Details

Particular	Details
Name of Student	Anuraj Kumar
Roll Number	25SCS1003001412
Batch	2CSE7
Programme	B.Tech (CSE – AI & ML)
University	IILM University, Greater Noida

Particular	Details
Company Name	Busung Tape & Film India Private Limited
Internship Type	Paid Internship
Duration	20 June 2026 – 10 July 2026
Stipend	6,000 (Rupees Six Thousand)
Certificate No.	BTC/INT/2026/02
Certificate Date	10 July 2026

1.2 Objectives of the Internship

The internship was undertaken with the following broad and specific objectives:

Broad Objectives

- To gain practical exposure to the functioning of a manufacturing organization engaged in tape and film production.
- To understand the interface between business functions (Sales, Marketing, Data Management) and production operations.
- To apply analytical and data-handling skills acquired during academic study to real industrial data and processes.
- To develop professional attributes such as discipline, communication, teamwork and documentation skills.

Specific Objectives

- To assist the Sales & Marketing team in day-to-day activities and related documentation.
- To perform Data Analysis and prepare structured reports using Microsoft Excel.
- To support Business Data Management tasks and maintain organized records.
- To study the circular die-cutting process and identify sources of material wastage.
- To design and help implement an optimized layout of six dies that converts side scrap into usable product.
- To quantify, to the extent possible, the reduction in wastage and the consequent impact on product cost.

- To document the entire process improvement activity in a clear and professional manner.

1.3 Scope and Limitations

Scope: The internship covered work at the Greater Noida manufacturing facility of Busung Tape & Film India Private Limited. The functional areas included Sales & Marketing support, Data Analysis & Reporting, Microsoft Excel based operations and Business Data Management. In addition, a focused process-improvement activity on the circular die-cutting line was undertaken. The period of engagement was strictly from 20 June 2026 to 10 July 2026.

Limitations: Being a short-duration internship of approximately three weeks, the depth of quantitative data collection was limited by the availability of historical production records and by the time required for trial implementation of the new die layout. Confidential commercial data (exact material costs, customer pricing, detailed production volumes) could not be reproduced in this report. The analysis therefore emphasizes the qualitative nature of the improvement and the directional impact on wastage and cost rather than precise rupee figures. The solution was developed and trialled within the constraints of existing dies and machinery; no capital modification of equipment was involved.

1.4 Organization of the Report

This report is organized into eleven chapters followed by Bibliography and Appendix. Chapter 1 introduces the background, objectives and scope. Chapter 2 presents a detailed profile of the host organization. Chapter 3 describes the internship role and day-to-day responsibilities. Chapter 4 analyses the material wastage problem observed in circular die cutting. Chapter 5 explains the proposed six-die optimized layout and its implementation. Chapter 6 covers the data analysis, Excel and business-data work performed. Chapters 7 and 8 discuss learning outcomes and challenges. Chapter 9 offers recommendations. Chapter 10 presents the conclusion and Chapter 11 outlines future scope. Supporting references and certificate details appear at the end.

2. ORGANIZATION PROFILE

2.1 About the Company

Busung Tape & Film India Private Limited is a private limited company incorporated in India and engaged in the manufacturing and processing of specialized tapes and films. The company is registered under the Companies Act with Corporate Identity Number (CIN) U74999UP2019FTC115758.

The organization operates in the industrial materials segment, supplying products that find application in packaging, insulation, electrical, automotive and general industrial uses. Precision in material handling, coating, laminating and converting (including die-cutting and slitting) is central to the company's value proposition. As a relatively young but growing manufacturing unit located in the industrial belt of Greater Noida, the company places importance on operational efficiency, quality consistency and cost competitiveness.

The company is led by Mr. Avinash Kumar (Director, DIN: 08488139). During the internship period, he served as the primary authorizing signatory for the Internship Offer Letter and the Internship Completion Certificate. Communication with the company is maintained through the official email busung.india@gmail.com and telephone number 0120-2341192.

2.2 Location and Infrastructure

The manufacturing facility is situated at:

F-5, Site 5, UPSIDA, Kasna, Surajpur, Greater Noida, Uttar Pradesh, India – 201306

(Also referred in the Offer Letter as Site-5, F5, Jail Road, Block F, Surajpur Site V, Greater Noida – 201312.)

The location falls within the industrial area developed by the Uttar Pradesh State Industrial Development Authority (UPSIDA). Proximity to the National Capital Region provides logistical advantages for both inbound raw-material movement and outbound finished-goods distribution. The facility houses equipment for slitting, laminating, coating and converting operations, along with associated storage, quality-check and administrative spaces.

2.3 Core Business Activities

The company's core technical competencies, as indicated on its official letterhead and certificates, are:

- Slitting – precision cutting of wide rolls into narrower widths as per customer specification.
- Laminating – bonding of multiple layers of film or tape materials to achieve desired functional properties.
- Coating – application of functional coatings (adhesive, release, protective, etc.) onto substrate films.

In addition to these primary processes, converting operations such as die-cutting (including circular die-cutting) are performed to produce finished shapes required by customers. Material utilization efficiency at every stage – especially during die-cutting – has a direct bearing on raw-material cost and overall product competitiveness. It was within this converting stage that the wastage-reduction opportunity was identified during the internship.

2.4 Organizational Structure (Overview)

Being a private limited manufacturing enterprise of moderate scale, the organization maintains a relatively flat structure. Key functional areas observed during the internship include:

- Direction / Top Management (Director)
- Production and Shop-floor Operations (Slitting, Laminating, Coating, Converting)
- Sales and Marketing
- Quality Control / Assurance
- Stores and Material Management
- Accounts and Administration / HR

Interns were primarily attached to the Sales & Marketing and data-support functions, with freedom of observation in the cross-functional exposure provided highly beneficial for understanding how commercial decisions and production efficiencies are interlinked.

2.5 Products and Market Presence

The company manufactures and supplies a range of industrial tapes and films. While exact product catalogues and customer lists are confidential, the general product categories include adhesive tapes, specialty films and converted (die-cut) components used by industrial customers. The market is competitive, and customers are sensitive both to quality consistency and to price. Any reduction in material wastage therefore translates into improved margins or more competitive pricing,

strengthening the company's position in the market.

Table 2: Company Profile Snapshot

Attribute	Information
Legal Name	Busung Tape & Film India Private Limited
CIN	U74999UP2019FTC115758
Registered / Factory Address	F-5, Site 5, UPSIDA, Kasna, Surajpur, Greater Noida, UP – 201306
Director	Mr. Avinash Kumar (DIN: 08488139)
Email	busung.india@gmail.com
Telephone	0120-2341192
Core Processes	Slitting, Laminating, Coating, Converting (Die-cutting)
Internship Certificate No.	BTC/INT/2026/02

3. INTERNSHIP ROLE AND RESPONSIBILITIES

3.1 Position Offered

As per the Internship Offer Letter dated 17 June 2026, I was offered the position of Intern – Sales & Marketing / Data Analysis. The offer was based on academic background and expressed interest in the areas of Sales & Marketing, Data Analysis and Microsoft Excel Operations. The internship was paid, with a stipend of 6,000 for the complete period.

The working location was the company premises at Site-5, Greater Noida. Working hours and attendance norms of the company were to be followed. The Offer Letter also contained clear Terms and Conditions covering confidentiality, adherence to company policies, mandatory satisfactory performance and attendance, ownership of work produced, and the company's right to terminate in case of misconduct or unsatisfactory performance. Upon successful completion, the company issued the Internship Completion Certificate.

3.2 Key Working Areas

The four primary working areas formally recognized in both the Offer Letter and the Completion Certificate are:

1. Sales & Marketing

Support to the sales team in preparation of basic sales-related documents, customer follow-up tracking, maintenance of enquiry and order status records, and assistance in marketing communication materials where required. Understanding customer requirements and aligning internal data accordingly.

2. Data Analysis & Reporting

Collection, cleaning and analysis of operational and sales-related data. Preparation of periodic summary reports that help supervisors take informed decisions. Identification of patterns or anomalies in the data that may indicate process or commercial issues.

3. Microsoft Excel Operations

Extensive use of Microsoft Excel for data entry, validation, formula-based calculations, pivot tables, conditional formatting, basic charts and structured report generation. Excel served as the primary tool for almost all analytical and reporting tasks.

4. Business Data Management

Organizing and maintaining business data in a systematic manner so that it remains accurate, accessible and useful for future reference. Ensuring version control and proper naming conventions for files and folders related to the internship work.

3.3 Daily Responsibilities

A typical day during the internship involved a combination of the following activities:

- Reporting to the assigned supervisor and reviewing pending tasks.
- Updating sales enquiry / order tracking sheets in Excel.
- Performing data cleaning and preparing short analytical summaries.
- Assisting in preparation of internal reports or presentations.
- Observing production processes (especially converting / die-cutting) when permitted.
- Documenting observations related to material wastage and layout ideas.
- Participating in brief discussions with production or sales personnel.
- Maintaining a daily work log for personal and reporting purposes.

Attendance and punctuality were strictly maintained, as these were mandatory conditions for successful completion of the internship and issuance of the certificate.

3.4 Tools and Software Used

The principal tool used throughout the internship was Microsoft Excel. Specific features employed included:

- Data entry and validation rules
- Formulas and functions (SUM, AVERAGE, IF, VLOOKUP / XLOOKUP style lookups, COUNTIF, etc.)
- Pivot Tables for summarization
- Conditional formatting for visual highlighting of exceptions
- Basic charts (bar, line, pie) for presentation of trends
- Sorting, filtering and data consolidation across multiple sheets

In addition, standard office tools for document preparation and the company's internal file-sharing / communication practices were followed. No specialized AI or advanced analytics platform was deployed during the short internship; the emphasis remained on practical, immediately usable Excel-based analysis.

Table 3: Key Working Areas and Sample Responsibilities

Working Area	Sample Responsibilities
Sales & Marketing	Enquiry tracking, order status updates, basic customer follow-up support
Data Analysis & Reporting	Cleaning data, preparing summary reports, identifying trends
Microsoft Excel Operations	Pivot tables, formulas, charts, structured workbooks
Business Data Management	Organizing files, maintaining master data, version control
Process Improvement (Additional)	Study of die-cutting waste, design of six-die layout

4. PROBLEM IDENTIFICATION AND ANALYSIS

4.1 Observation of Production Process

During the early days of the internship, after becoming familiar with the office and data-related tasks, I was given the opportunity to visit the production floor and observe the converting operations. One of the processes that attracted attention was circular die-cutting. In this process, circular dies are used to cut round or disc-shaped products from continuous film or tape material.

The material typically arrives in roll form. It is fed into a die-cutting station where one or more circular dies cut the required shapes. After cutting, the finished circular pieces are separated, and the remaining material (the “skeleton” or side scrap) is usually collected and discarded or sent for recycling at low value. The geometry of circular cuts inevitably leaves interstitial material between the circles; the efficiency of material usage therefore depends heavily on how the dies are arranged relative to one another and relative to the width of the incoming web.

4.2 Material Wastage in Circular Die Cutting

It was observed that a noticeable quantity of side scrap was being generated and discarded after each production run. When the dies were arranged in a conventional (non-optimized) manner, the gaps between adjacent circular cuts and the margins at the edges of the web remained unutilized. Over successive shifts, this continuous generation of scrap translated into:

- Higher consumption of raw material per unit of finished product.
- Increased direct material cost of the final product.
- Greater volume of scrap that requires handling and disposal.
- Lower overall material yield (usable output / input material).

Because the raw materials used for specialty tapes and films are relatively expensive, even a modest percentage improvement in material utilization can produce meaningful cost savings and improve the competitiveness of the product.

4.3 Root Cause Analysis

The root cause of the excessive side scrap was identified as the sub-optimal spatial arrangement of the circular dies. In a conventional layout, dies are often placed in a simple rectangular or lightly staggered grid. For circular shapes, a rectangular grid leaves relatively large unusable regions between the circles. A more compact nesting (for example, hexagonal or carefully calculated staggered placement) can significantly reduce the interstitial waste.

Additional contributing factors that were noted conceptually include:

- Lack of a formal layout-optimization study before setting up the die station for a new job.
- Priority given to ease of setup and machine speed over maximum material utilization.
- Limited time available on the shop floor for experimental trials of alternative layouts.
- Absence of simple visual or quantitative guidelines for operators on optimal die spacing.

The internship provided an opportunity to focus specifically on the layout aspect and to propose a practical arrangement of six dies that would convert a substantial portion of the previous side scrap into additional usable product.

4.4 Impact of Wastage on Cost and Efficiency

Material cost typically constitutes a major share of the total cost of tape and film products. When a significant percentage of the purchased material is lost as scrap, the effective material cost per good unit rises. This either erodes the manufacturer's margin or forces a higher selling price, both of which are undesirable in a competitive market.

From an operational-efficiency standpoint, high scrap rates also mean more frequent material changeovers, more scrap handling labour, and greater environmental burden associated with disposal or low-value recycling. Therefore, any improvement that increases the number of good pieces obtained from the same length or area of input material delivers benefits on cost, productivity and sustainability dimensions simultaneously.

The problem was therefore framed as follows: "How can the arrangement of circular dies be modified so that a larger fraction of the input material is converted into saleable product, without requiring new capital equipment or changing the product specification?"

5. PROPOSED SOLUTION AND IMPLEMENTATION

5.1 Concept of Optimized Die Layout

The central idea of the solution was to move from a conventional, loosely spaced die arrangement to a more compact, carefully planned layout of six circular dies. By optimizing the relative positions of the six dies, the interstitial gaps that previously became side scrap could be reduced, and a meaningful portion of that material could be converted into additional usable circular pieces.

The optimization did not involve changing the diameter of the individual dies or the technical specification of the finished product. It focused purely on geometry – how the existing (or similarly sized) dies could be placed relative to one another and relative to the web so that material utilization improved. This approach has the advantage of being low-cost and quickly implementable.

5.2 Design of Six-Die Arrangement

After studying the available web width, the diameter of the circular product and the mechanical constraints of the die-cutting station, a layout using six dies was conceptualized. The arrangement aimed to:

- Maximize the number of complete circular cuts per unit length of material.
- Minimize the width of unusable edge margins and inter-die gaps.
- Maintain adequate mechanical clearance so that the dies could be mounted safely and operated at normal production speed.
- Remain simple enough for operators to set up repeatedly without specialized metrology equipment.

The six-die configuration was chosen because it offered a practical balance between complexity of setup and potential material saving for the typical order sizes and web widths handled by the company during the internship period.

5.3 Implementation Steps

The implementation followed a logical sequence:

Step 1 – Baseline Observation: Documented the existing die arrangement and qualitatively assessed the volume of side scrap generated.

Step 2 – Layout Design: Prepared a conceptual sketch of the six-die optimized arrangement, taking into account web width and die diameter.

Step 3 – Discussion with Production Team: Presented the idea to the relevant supervisors and operators, incorporated practical feedback on mounting and

safety clearances.

Step 4 – Trial Setup: With the assistance of the production staff, the six dies were arranged according to the proposed layout on a trial basis.

Step 5 – Trial Run: A limited quantity of material was processed under the new layout. The resulting scrap and usable output were observed and compared with the previous practice.

Step 6 – Evaluation: Confirmed that a significant portion of the material that previously became side scrap was now being converted into usable product. No adverse effect on product quality or machine stability was reported.

Step 7 – Documentation: Recorded the before-and-after observations and prepared a short internal note summarizing the improvement.

5.4 Results Achieved

The principal results of the intervention were:

- Reduction in raw-material wastage: A significant portion of the side scrap that was earlier discarded was converted into usable product output.
- Lower product cost: Because more good pieces were obtained from the same input material, the effective material cost per finished unit decreased.
- Improved material yield: The ratio of usable output to input material improved under the optimized six-die layout.
- No capital investment: The improvement was achieved solely through better arrangement of existing dies; no new machinery was required.
- Operator acceptance: After initial explanation, the production team was able to adopt the new layout for the relevant jobs.

While exact percentage figures for waste reduction are not reproduced in this report (owing to confidentiality of production data), the directional improvement was clear and was acknowledged by the supervisors concerned. The experience demonstrated that even a short internship can yield a tangible process improvement when the intern combines careful observation with structured problem-solving.

Table 4: Conceptual Comparison of Material Utilization

Parameter	Conventional Layout	Optimized Six-Die Layout
Number of dies considered	Lower / non-optimized	Six dies in compact arrangement
Inter-die gaps	Relatively large	Minimized through better nesting
Side scrap generation	High	Significantly reduced

Parameter	Conventional Layout	Optimized Six-Die Layout
Usable output from same material	Baseline	Higher
Impact on product cost	Higher material cost component	Lower material cost component
Capital requirement	—	None (layout change only)

6. DATA ANALYSIS AND REPORTING WORK

6.1 Sales and Marketing Support

In parallel with the process-improvement activity, a substantial portion of the internship time was devoted to supporting the Sales and Marketing function.

Typical tasks included:

- Updating enquiry registers and order-status trackers in Excel.
- Assisting in the preparation of simple customer-wise or product-wise summary statements.
- Helping maintain follow-up lists so that pending quotations or orders could be monitored.
- Organizing soft copies of sales-related documents for easy retrieval.
- Supporting the team with basic data required for internal discussions or customer communication.

Through these activities I gained an appreciation of how sales information flows within a manufacturing company and how timely, accurate data supports better commercial decisions.

6.2 Microsoft Excel Operations

Microsoft Excel was the workhorse tool for almost all analytical and reporting assignments. Specific operations performed included:

- Designing clean, well-structured worksheets with appropriate headers and data-validation rules.
- Using formulas to calculate totals, averages, differences and simple ratios.
- Building Pivot Tables to summarize large data sets by product, customer, date or status.
- Applying filters and conditional formatting to highlight exceptions or priority items.
- Creating basic charts to visualize trends for internal review.
- Consolidating data from multiple sheets or sources into master reports.

These exercises reinforced classroom knowledge of spreadsheet software and demonstrated how even intermediate Excel skills can deliver immediate value in a business setting.

6.3 Business Data Management

Good data management practices were emphasized throughout the internship.

Responsibilities in this area included:

- Maintaining consistent file-naming conventions and folder structures.
- Ensuring that updated versions of trackers and reports replaced older ones in a controlled manner.
- Protecting the confidentiality of company and customer data as required by the Terms and Conditions of the internship.
- Preparing handover notes so that the work done during the internship remained usable after the internship ended.

The combination of Sales support, Excel-based analysis and disciplined data management formed the core of the formal internship role, while the die-layout improvement represented an additional, self-initiated contribution that the company welcomed.

Table 5: Tools and Software Used

Tool / Software	Purpose during Internship
Microsoft Excel	Data entry, analysis, Pivot Tables, charts, reporting
MS Office (Word / basic docs)	Preparation of notes, summaries and formal documentation
Company internal trackers	Enquiry and order status maintenance
Manual observation & measurement	Study of die layout and scrap generation on shop floor

7. LEARNING OUTCOMES AND SKILLS DEVELOPED

The internship, although of short duration, produced a rich set of learning outcomes that complement the academic curriculum of B.Tech (CSE – AI & ML). These outcomes may be grouped under technical, analytical, process-oriented and professional categories.

7.1 Technical and Analytical Skills

- Strengthened proficiency in Microsoft Excel for real business data (beyond classroom exercises).
- Learned to clean, structure and summarize operational and sales data for decision support.
- Gained first-hand understanding of how geometric layout decisions affect material yield in converting processes.
- Practised systematic problem identification, root-cause thinking and solution design in an industrial context.
- Understood the linkage between production efficiency and product cost.

7.2 Process and Domain Knowledge

- Acquired basic familiarity with tape and film manufacturing processes – especially slitting, laminating, coating and die-cutting.
- Understood the importance of material utilization and scrap reduction in a cost-sensitive manufacturing environment.
- Observed how Sales, Production and Data functions interact in a medium-sized manufacturing company.
- Learned the practical constraints (machine limitations, setup time, operator convenience) that must be respected when proposing process changes.

7.3 Professional and Soft Skills

- Developed the habit of punctuality, regular attendance and adherence to organizational rules.
- Improved communication with supervisors, operators and cross-functional colleagues.
- Learned to document work clearly so that others can understand and continue it.
- Experienced the discipline of confidentiality and professional ethics required in a company setting.

- Gained confidence in presenting ideas (such as the six-die layout) and incorporating feedback.

7.4 Alignment with AI & ML Academic Background

Although the internship did not involve coding of machine-learning models, the analytical mindset fostered by the AI & ML curriculum proved useful. The ability to observe a process, collect relevant data (even if limited), identify patterns of waste, and propose a structured improvement is closely related to the data-driven problem-solving approach emphasized in Artificial Intelligence and Machine Learning studies. In future roles, the same approach can be extended with sensor data, statistical process control or optimization algorithms.

8. CHALLENGES FACED AND HOW THEY WERE OVERCOME

Several challenges were encountered during the internship. Each challenge and the manner in which it was addressed is described below.

8.1 Limited Duration of Internship

Challenge: A three-week internship is relatively short for deep process studies or large-scale data analysis projects.

How overcome: Prioritized one high-impact, low-complexity improvement (die-layout optimization) that could be observed, designed and trialled within the available time. Parallel office tasks were managed through disciplined daily planning.

8.2 Restricted Access to Detailed Production Data

Challenge: Exact production volumes, scrap percentages and material cost figures are commercially sensitive and were not available for unrestricted use in the report.

How overcome: Focused on qualitative and directional results. Described the nature of the improvement and its logical impact on wastage and cost without disclosing confidential numbers. Obtained verbal confirmation from supervisors regarding the usefulness of the change.

8.3 Balancing Office Work and Shop-floor Observation

Challenge: Primary responsibilities were in Sales / Data / Excel support; shop-floor access was secondary and depended on production schedules.

How overcome: Coordinated with supervisors to schedule observation time without disrupting production. Used available windows efficiently and documented observations immediately.

8.4 Communicating Technical Ideas to Shop-floor Personnel

Challenge: Explaining a geometric layout concept to operators and supervisors who are focused on daily output targets required clear, non-academic communication.

How overcome: Used simple sketches and practical language. Emphasized the benefit (less scrap, more good pieces from same material) rather than theoretical optimization theory. Incorporated their feedback on mounting practicality.

8.5 Maintaining Confidentiality and Professional Boundaries

Challenge: The Terms and Conditions of the internship required strict confidentiality of company information.

How overcome: Consciously avoided recording or reproducing any commercially sensitive figures, customer names or proprietary process parameters in personal notes or in this report. Shared only what was necessary for academic documentation and had been cleared in principle by the nature of the certificate issued.

9. RECOMMENDATIONS

Based on the observations and work performed during the internship, the following recommendations are offered for the consideration of the company and for future interns / students.

9.1 Recommendations for the Company

- Standardize optimized die layouts: For recurring circular products, document and display the preferred six-die (or other optimized) arrangements so that operators can set them up consistently.
- Simple material-yield tracking: Introduce a basic Excel or shop-floor log that records input material length/area and number of good pieces produced. Even a simple yield percentage can highlight opportunities for further improvement.
- Short Kaizen-style trials: Encourage small, low-cost layout or process trials and recognize successful ideas, regardless of whether they originate from permanent staff or interns.
- Cross-functional exposure for interns: Continue the practice of allowing interns to observe production in addition to their primary data/sales roles; this broadens their contribution potential.
- Retain internship documentation: Keep a short internal note of process improvements suggested by interns so that institutional knowledge is not lost when the internship ends.

9.2 Recommendations for Future Interns / Students

- Treat even a short internship as an opportunity to deliver at least one tangible improvement, however small.
- Combine assigned office/data work with careful observation of physical processes; the most valuable insights often lie at the interface of data and operations.
- Document everything – daily logs, sketches, before-and-after observations – while the details are fresh.
- Respect confidentiality and company culture at all times; trust is the foundation of meaningful access.
- Use Excel (or similar tools) rigorously; clean data and clear presentation multiply the impact of any analysis.

9.3 Recommendations for the Academic Institution

- Continue to emphasize internships that allow students to work on real problems rather than purely observational assignments.
- Encourage students to include a clear problem–solution–result narrative in their internship reports, as this develops both technical and communication skills.
- Where possible, facilitate slightly longer internship windows so that quantitative validation of improvements can be more robust.

10. CONCLUSION

The paid internship at Busung Tape & Film India Private Limited from 20 June 2026 to 10 July 2026 provided a valuable opportunity to apply academic knowledge in a real manufacturing and business environment. Working in the areas of Sales & Marketing, Data Analysis & Reporting, Microsoft Excel Operations and Business Data Management gave me practical experience in supporting commercial and operational decision-making through organized data work.

Beyond the formally assigned responsibilities, the identification of material wastage in the circular die-cutting process and the subsequent implementation of an optimized six-die layout constituted a concrete contribution. By converting a significant portion of previously discarded side scrap into usable product, the intervention helped reduce raw-material wastage and lower the overall product cost. The improvement required no capital investment – only better geometric arrangement of existing dies – demonstrating that careful observation and structured thinking can yield measurable benefits even in a short internship.

The company formally recognized the successful completion of the internship through Certificate No. BTC/INT/2026/02 dated 10 July 2026, and noted that the intern demonstrated dedication, sincerity, willingness to learn and professionalism. Performance was assessed as satisfactory.

Overall, the internship strengthened technical, analytical and professional skills, deepened understanding of manufacturing process efficiency, and reinforced the importance of data-driven and observation-driven problem solving – competencies that are highly relevant to a future career in Computer Science, Artificial Intelligence, Machine Learning or industrial analytics.

Table 6: Summary of Key Achievements

Achievement Area	Description
Process Improvement	Optimized six-die layout reducing side scrap in circular die-cutting
Cost Impact	Lowered effective raw-material cost per finished product
Data & Reporting	Supported Sales/Marketing with Excel-based analysis and trackers
Professional Conduct	Maintained attendance, confidentiality and teamwork standards
Certification	Received formal Internship Completion Certificate (BTC/INT/2026/02)

11. FUTURE SCOPE

The work begun during the internship can be extended in several directions, both within the host company and in the broader context of industrial process optimization and data analytics.

11.1 Within the Company

- Extend the layout-optimization approach to other product sizes and to non-circular die shapes.
- Develop a simple visual standard or job-aid for operators showing preferred die arrangements for common products.
- Introduce basic digital recording of material input versus good output so that yield trends can be monitored over time.
- Explore whether low-cost sensors or vision systems could later assist in automatic scrap detection or layout verification.

11.2 Academic and Career Perspective

- Apply more advanced optimization techniques (for example, mathematical nesting algorithms or simulation) to die-layout problems in future projects or higher studies.
- Combine process data with statistical or machine-learning methods to predict scrap rates or recommend optimal setups automatically.
- Pursue further internships or projects that sit at the intersection of manufacturing operations and data science / AI.
- Build a personal portfolio of documented process-improvement case studies that demonstrate practical impact.

The internship has thus served not only as a fulfilment of academic requirements but also as a foundation for continued learning in the domains of industrial efficiency and data-supported decision making.

BIBLIOGRAPHY / REFERENCES

- Busung Tape & Film India Private Limited. (2026). Internship Offer Letter dated 17 June 2026. Greater Noida, Uttar Pradesh.
- Busung Tape & Film India Private Limited. (2026). Internship Completion Certificate, Certificate No. BTC/INT/2026/02, dated 10 July 2026. Greater Noida, Uttar Pradesh.
- Company internal process observations, production-floor discussions and data shared during the internship period (20 June 2026 – 10 July 2026). (Confidential; not publicly citable).
- IILM University, Greater Noida – School of Computer Science and Engineering. Guidelines for Internship Report Format (as provided to students).
- Microsoft Corporation. Microsoft Excel documentation and standard data-analysis practices applied during reporting tasks.
- General principles of lean manufacturing and material utilization in converting industries (conceptual reference only; no specific copyrighted text reproduced).
- Personal daily work logs and sketches prepared by the intern during the internship period.

APPENDIX

Certificate and Offer Letter – Key Extracted Details

The original Internship Offer Letter (dated 17 June 2026) and Internship Completion Certificate (Certificate No. BTC/INT/2026/02, dated 10 July 2026) are to be attached with the hard copy of this report. For reference, the essential particulars extracted from these documents are reproduced below.

A. From Internship Offer Letter

- Date: 17 June 2026
- Addressee: Mr. Anuraj Kumar, Roll No. 25SCS1003001412, IILM University, Greater Noida
- Subject: Offer of Paid Internship
- Position: Intern – Sales & Marketing / Data Analysis
- Company: Busung Tape & Film India Private Limited
- Location: Site-5, F5, Jail Road, Block F, Surajpur Site V, Greater Noida, UP – 201312
- Internship Period: 20/06/2026 to 10/07/2026
- Stipend: 6,000 (Rupees Six Thousand Only)
- Working Areas: Sales & Marketing Activities; Data Analysis and Reporting; Microsoft Excel Operations; Business Data Management
- Terms: Confidentiality, adherence to policies, attendance & performance mandatory, company ownership of work produced, right to terminate for misconduct / unsatisfactory performance
- Signatory: Mr. Avinash Kumar, Director (DIN: 08488139)

B. From Internship Completion Certificate

- Certificate No.: BTC/INT/2026/02
- Date: 10 July 2026
- Intern: Mr. Anuraj Kumar, Roll No. 25SCS1003001412
- Programme: B.Tech (CSE – AI & ML), IILM University, Greater Noida
- Company: Busung Tape & Film India Private Limited
- Period: 20 June 2026 to 10 July 2026
- Nature: Paid Internship successfully completed
- Areas of Work: Sales & Marketing; Data Analysis & Reporting; Microsoft Excel Operations; Business Data Management

- Performance Remark: Demonstrated dedication, sincerity, willingness to learn and professionalism; performance satisfactory; contribution appreciated
 - Authorized Signatory: (HR / Director), Busung Tape & Film India Private Limited
-

— End of Internship Report —

*Anuraj Kumar / Roll No. 25SCS1003001412 / Batch 2CSE7 / B.Tech (CSE – AI & ML)
IILM University, Greater Noida*

ADDITIONAL NOTES ON PROCESS IMPROVEMENT

Significance of Layout Optimization in Converting Industry

In the converting industry, material utilization is one of the most critical drivers of profitability. Unlike continuous process industries where yield losses are often chemical or thermal in nature, converting operations lose material primarily through geometric inefficiency. Circular or irregular shapes cut from a continuous web inevitably leave residual material. The percentage of this residual material that can be recovered or minimized depends almost entirely on how intelligently the cutting tools (dies) are arranged.

The six-die arrangement implemented during the internship is a practical illustration of this principle. By treating die placement as a geometric optimization problem rather than a purely mechanical setup task, it was possible to increase the number of good pieces obtained from the same length of material. This approach aligns with lean manufacturing philosophy, which emphasizes the elimination of all forms of waste (muda), including material waste.

For a student of Computer Science and Engineering with specialization in Artificial Intelligence and Machine Learning, the experience also highlighted a future opportunity: the same layout problem can be formalized as a mathematical optimization or constraint-satisfaction problem and solved algorithmically for more complex product mixes. Sensor data and computer vision could further automate the verification of correct die placement and real-time scrap monitoring.

Personal Reflection

Looking back on the three-week internship, the most valuable lesson was that meaningful contribution does not always require advanced technology or large datasets. Careful observation, structured thinking and clear communication with the people who run the process every day can produce improvements that benefit the organization immediately. The formal recognition through the Internship Completion Certificate is appreciated, but the deeper satisfaction comes from knowing that a practical idea moved from observation to trial implementation within a short span of time.

I remain grateful to Busung Tape & Film India Private Limited and to IILM University for making this learning experience possible. I look forward to applying the same spirit of observation and continuous improvement in future academic projects and professional roles.

— End of Additional Notes —